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Prescribed-time adaptive constraint control of an uncertain nonlinear 2-DOF helicopter

Vijay Kumar Singh * **Shyam Kamal** * **Ankit Sachan** **
Sandip Ghosh *

* *Department of Electrical Engineering, Indian Institute of Technology (BHU), Varanasi, India (e-mail: vijaykumarsingh.rs.eee19@iitbhu.ac.in, shyamkamal.eee@iitbhu.ac.in, sghosh.eee@iitbhu.ac.in).*
** *Graduate School of Advanced Science and Engineering, Hiroshima University, Higashi Hiroshima, Japan (e-mail: sachan@hiroshima-u.ac.jp).*

Abstract: This paper addresses the problem of attitude tracking for a 2-Degree of Freedom (DOF) helicopter system with uncertainties. Euler-Lagrange equations are utilized to develop a nonlinear model for the helicopter, considering the control input constant coefficients and system parameters as unknown. The prescribed-time control approach is proposed using adaptive backstepping to track the desired pitch and yaw positions separately. In Addition, a constrained function is also used to avoid the risk of a larger signal when time gets closer to the appointed settling time. Through the application of the theory of Lyapunov stability, it is shown that the proposed strategy ensures that all closed-loop signals are bounded, and all the states converge within the prescribed time. A significant advantage of this approach is the ability to pre-specify the upper bound of the settling time. Finally, the efficacy and control capabilities of the proposed scheme are verified by obtaining the simulation results on the Quanser 2-DOF helicopter.

Keywords: 2-DOF helicopter, adaptive control, backstepping, prescribed-time stabilization, trajectory tracking.

1. INTRODUCTION

Over the past few decades, unmanned aerial vehicles and helicopters have gained popularity in various fields, including transportation, military, search and rescue, law enforcement, and aerial photography. However, in order to perform challenging maneuvers under challenging flight conditions while remaining stable, reliable, and safe, helicopters require advanced control systems. Due to model uncertainties, nonlinear behavior, cross-coupling, and external disturbances, significant research has been conducted to control such systems. Designing a flight control system is critical in creating a fully operating helicopter, and various control techniques have been successfully applied in the literature, including traditional PID control (Boubakir et al. (2013)), linear quadratic regulator (LQR) control (Pandey et al. (2015)), and more sophisticated controllers such as neural network approach, differential geometry method, robust composite nonlinear feedback control with decoupling approach, model predictive control (MPC), resilient sliding mode control, and nonlinear H_∞ control. Additionally, identifying helicopter models is not a simple or quick task, often requiring specific experiments to be carried out numerous times to obtain more detailed findings due to the intrinsic complexity of these models (Marconi et al. (2008); Chen et al. (2015); Joelianto et al. (2011)).

Controlling helicopters is challenging due to their vulnerability to internal and external uncertainty, such as operating in windy conditions (Åström (1983)). In some cases, it may be necessary to control an unknown plant. Adaptive control is a well-known and crucial method for dealing with system uncertainties, as it offers online estimations of ill-known system parameters using measurements. The backstepping technique has been extensively employed to develop adaptive controllers for uncertain systems, offering a means of enhancing the transient performance of adaptive systems over conventional methods. As a result, backstepping-based adaptive control of helicopters has drawn significant attention (Zou et al. (2015); Wang et al. (2021); Zhang et al. (2022); Al Issa et al. (2022)). Recent developments in adaptive control of helicopter systems can be found in literature such as (Zhou et al. (2004)) and (Krstic et al. (1995)). While (Zhou et al. (2004)) developed a simple adaptive control for a quadrotor helicopter that experienced control effectiveness reduction, they considered a linearized system, which simplified the controller design. Similarly, a linear helicopter system was devised for model reference adaptive control (MRAC) in (Wen et al. (2011)).

Many researchers have widely studied the rated convergence of the 2-DOF helicopter system. However, it is important to note that in cases of finite-time convergence,

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the upper bound of the convergence time is not uniform with respect to the initial conditions (Bhat et al. (2000)). Therefore, estimating the convergence time is difficult if the initial conditions are unknown, which is often the case for practical systems. Furthermore, if the initial conditions are far from the equilibrium point, the convergence time can be very long, which can negatively affect the overall system performance (Ding et al. (2021); Adetola et al. (2008)). Fortunately, the concept of fixed-time stability, proposed in (Polyakov (2011)), provides a new direction for limited-time convergence. This concept retains all the benefits of control based on finite-time stability, and the settling time is uniform regardless of the magnitude of the initial conditions, meaning that convergence can be achieved in a fixed time regardless of the initial conditions. Due to the excellent benefits of control based on fixed-time stability, many researchers have applied it to various stabilization and tracking control problems (Chatterjee et al. (2013)).

While previous works have established the convergence of system states in a fixed time frame, determining the relationship between the controller's design parameters and the convergence time-bound remains challenging. Furthermore, the upper bound of convergence time cannot be predetermined. However, a class of stabilization techniques called prescribed-time stability and stabilization enables one to define the upper bound of convergence time in advance (Song et al. (2017); Pal et al. (2020)). These time-varying methodologies have surpassed the limitations of controllers relying on fixed-time stabilization. Control design based on prescribed-time convergence has introduced a new dimension in the field of rated convergence (Singh et al. (2022)). However, these stabilization approaches are not applicable to systems containing unknown parameters, which are prevalent in modern complex systems. Hence, exploring prescribed-time stabilization techniques for systems with unknown constant parameters is of utmost importance (Singh et al. (2021)).

Based on the above observations, this article presents a novel adaptive prescribed-time constraint feedback control methodology for a 2-DOF helicopter systems with unknown constant parameters. The proposed approach employs adaptive laws to estimate the unknown parameters of the system and employs an adaptive backstepping technique to design the prescribed-time control law that ensures reference tracking of the system states within a predetermined time frame. The constraint function is also incorporated to restrict the virtual control signals. The boundedness of all signals of the closed-loop system is established using the Lyapunov's direct stability method. Results show that the proposed method outperforms existing methods in terms of convergence time and guarantees the boundedness of all signals of the closed-loop system.

This article is structured as follows. Section II provides preliminaries that will be useful for obtaining the main results. In Section III, the main results are shown. Section IV validate the efficacy of the proposed scheme with a practical example. Finally, Section V concludes this work.

2. PROBLEM STATEMENT

Let $\mathbb{R}$ stand for the set of all real numbers and $\mathbb{R}_{\geq 0}$ for the field of all real numbers that are not negative. The n-dimensional vector space over the field $\mathbb{R}$ is known as $\mathbb{R}^n$. The the Euclidean norm of a finite-dimensional vector is denoted by $\|\cdot\|$, and the absolute value of a scalar variable is represented by $\lfloor \cdot \rfloor$. The transpose of a vector or matrix is indicated by $(\cdot)^\top$.

2.1 On prescribed-time convergence

The following results will be useful in order to understand the concept of convergence in a given time, where the upper bound of the settling-time can be prescribed.

Let us consider the following forced nonlinear dynamical system

$$\dot{z} = \mathcal{F}(t, z, \Phi, u), \quad z(0) = z_0 \tag{1}$$

where, $z \in \mathbb{R}^n$ is the state vector, $\Phi \in \mathbb{R}^p$ is the uncertain parameter $u \in \mathbb{R}^m$ is the control input and $\mathcal{F} : \mathbb{R}_{\geq 0} \times \mathbb{R}^n \times \mathbb{R}^p \times \mathbb{R}^m \to \mathbb{R}^n$ is a nonlinear function and satisfies Lipschitz condition for z with $\mathcal{F}(t, 0, \Phi, 0) = 0$, and is piecewise continuous with respect to t. Then, the system (1) has a unique solution for all $z_0 \in \mathbb{R}^n$.

Definition 1. (Hua et al. (2021)) For the system (1), if there exists a control law $u = u(t, z, \hat{\Phi})$ with the parameter adaptive law $\dot{\hat{\Phi}} = \psi(t, z, \hat{\Phi})$, where $\hat{\Phi}$ is the estimate of Φ, such that for any bounded $z(0) \in \mathbb{R}^n$, the trajectories of z and $\hat{\Phi}$ are bounded and $z = 0 \; \forall t \geq T_p$, then the origin of system (1) is said to be prescribed-time stable with in a priori chosen time T_p.

Lemma 1. (Hua et al. (2021)) For the system (1), if there exist two positive definite continuous and differentiable functions $V_1(z)$, $V_2(\tilde{\Phi})$ and class $\mathcal{K}_\infty$ functions β_1, β_2, β_3 and β_4 such that $\forall z \in \mathbb{R}^n$, we have

$$V = V_1(z) + V_2(\tilde{\Phi}) \tag{2}$$

$$\beta_1(\|z\|) \leq V_1(z) \leq \beta_2(\|z\|) \tag{3}$$

$$\beta_3(\|\tilde{\Phi}|) \leq V_2(\tilde{\Phi}) \leq \beta_4(\|\tilde{\Phi}\|) \tag{4}$$

$$\dot{V} \leq \begin{cases} -\gamma\varphi(t, T_p)V_1(z), & \forall t \in [0, T_p) \\ 0, & \forall t \in [T_p, \infty) \end{cases} \tag{5}$$

where $\gamma > 1$ is a design constant and $\varphi(t, T_p)$ is the prescribed time adjustment function. Then the origin of the system (1) is prescribed-time stable and all the closed-loop signals are bounded.

Remark 1. It is worth mentioning that prescribed time adjustment function $\varphi(t, T_p)$ has several options with various convergence responses. For example:

$$\varphi(t, T_p) = \frac{1}{T_p - t}, \quad \varphi(t, T_p) = e^{\frac{T_p}{T_p - t}} - 1.$$

2.2 2-DOF helicopter model

The Quanser 2 DOF helicopter model is considered here. The free body diagram is shown in Fig. 2. With the help of the Euler-Lagrangian formula, the mathematical modeling obtained as follows:

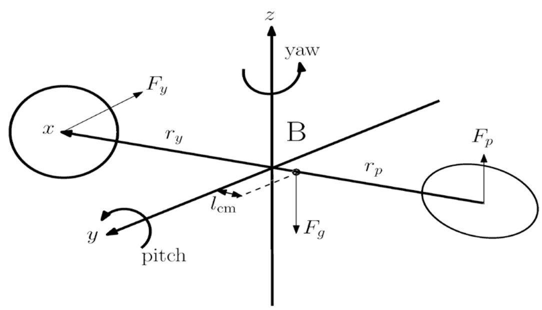

Fig. 1. Free-body diagram of 2-DOF helicopter model

$$\ddot{\theta} = -\frac{B_p\dot{\theta}}{J_p + m_h l^2} - \frac{m_h l^2 \dot{\psi}^2 \sin(\theta)\cos(\theta)}{J_p + m_h l^2} - \frac{m_h g l \cos(\theta)}{J_p + m_h l^2} + \frac{K_{pp}V_p}{J_p + m_h l^2} + \frac{K_{py}V_y}{J_p + m_h l^2} \tag{6}$$

$$\ddot{\psi} = \frac{2m_h l^2 \dot{\psi} \sin(\theta)\cos(\theta)\dot{\theta}}{J_y} - \frac{B_y\dot{\psi}}{J_y} + \frac{K_{yp}V_p}{J_y} + \frac{K_{yy}V_y}{J_y} \tag{7}$$

where, $\theta, \dot{\theta}, \psi$ and $\dot{\psi}$ denote the pitch angle, pitch angular velocity, yaw angle and yaw angular velocity respectively. V_p and V_y are the input voltages to the pitch and yaw motors.

Now, the state variables and control inputs are defined as

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \\ \psi \\ \dot{\psi} \end{bmatrix}, \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} u_p \\ u_y \end{bmatrix} \tag{8}$$

where, u_p and u_y are the pitch and yaw motors control inputs, and they are specified as

$$u_p = K_{pp}V_p + K_{py}V_y \tag{9}$$

$$u_y = K_{yp}V_p + K_{yy}V_y \tag{10}$$

Rewriting equations (6) and (7) into the state space form, we get

$$\begin{aligned} \dot{z}_1 &= z_2 \\ \dot{z}_2 &= \Theta_1^\top(z)\Phi_1 + b_p h_p(u_p) \\ \dot{z}_3 &= z_4 \\ \dot{z}_4 &= \Theta_2^\top(z)\Phi_2 + b_y h_y(u_y) \end{aligned} \tag{11}$$

where the known nonlinear functions Θ_1 and Θ_2 are defined as

$$\Theta_1 = \begin{bmatrix} -z_2 \\ -\cos(z_1) \\ -z_4^2\cos(z_1)\sin(z_1) \end{bmatrix}, \Theta_2 = \begin{bmatrix} -z_4 \\ z_2 z_4 \cos(z_1)\sin(z_1) \end{bmatrix}$$

Additionally, the unknown constant vectors Φ_1 and Φ_2 are specified as

$$\Phi_1 = \frac{1}{J_p + m_h l^2}\begin{bmatrix} Bp \\ m_h g l \\ m_h l^2 \end{bmatrix}, \quad \Phi_2 = \frac{1}{J_y}\begin{bmatrix} B_y \\ 2m_h l^2 \end{bmatrix}$$

and the unknown input coefficients b_p and b_y are defined as

$$b_p = \frac{1}{J_p + m_h l^2}, \quad b_y = \frac{1}{J_y}$$

Since, the aim is to develop the control law for u_1 and u_2 so that the outputs z_1 and z_3 follow the desired reference signals z_{d1} and z_{d3} respectively in a prescribed-time, which can be specified in advance. The following assumptions are made in order to attain the control objective.

Assumption 1: Φ_1, Φ_2, b_p, and b_y are all positive constants that represent the unknown constant parameters of the considered system.

Assumption 2: The first-order and second-order derivative of the desired reference signals are known, bounded and piecewise continuous.

3. PRESCRIBED-TIME ADAPTIVE CONSTRAINT FEEDBACK CONTROL DESIGN

For the considered 2-DOF helicopter system with uncertain parameters, we will use an adaptive backstepping technique to develop a prescribed-time feedback control law. We begin by applying the change of coordinates for the state variables as

$$x_1 = z_1 - z_{d1} \tag{12}$$

$$x_2 = z_2 - h_1(\sigma_1) - \dot{z}_{d1} \tag{13}$$

$$x_3 = z_3 - z_{d3} \tag{14}$$

$$x_4 = z_4 - h_2(\sigma_2) - \dot{z}_{d3} \tag{15}$$

where, σ_1 and σ_2 are the virtual controllers and chosen as

$$\sigma_1 = -\gamma_1\varphi(t, T_p)x_1$$

$$\sigma_2 = -\gamma_3\varphi(t, T_p)x_3$$

where, γ_1 and γ_3 are positive design constants.

Remark 2. Please note that the time-varying function $\varphi(t, T_p)$ might cause infinite control gain as $t \to T_p$. Also, there are restricted regions in the system states for practical application. Since we know that in backstepping technique, z_{i+1} considered a control input for the $i-th$ subsystem. If the desired virtual controllers are larger than the constrained region, it's possible that z_{i+1} struggles to track it. In order to address the previous two issues, we incorporate the following function to limit the control signals:

$$h_i(\sigma) = q_i^* \tanh\left(\frac{\sigma}{q_i^*}\right) = q_i^* \frac{e^{\sigma/q_i^*} - e^{-\sigma/q_i^*}}{e^{\sigma/q_i^*} + e^{-\sigma/q_i^*}} \tag{16}$$

where q_i^* is a positive constant. It is easy to show $h_i(\sigma)$ is a smooth bounded function which belongs to $(-q_i^*, q_i^*)$. Moreover, $h_i(\sigma)$ is differentiable at $\sigma = 0$ and can be transformed into the following equation by the mean-value theorem:

$$h_i(\sigma) = \frac{\partial h_i(\bar{\sigma})}{\partial \sigma}(\sigma - \sigma_0) + h_i(\sigma_0) \tag{17}$$

where $\bar{\sigma} \in (\sigma - \sigma_0)$, $\sigma_0 < \sigma$, and

$$\frac{\partial h_i(\bar{\sigma})}{\partial \sigma} = \frac{4}{(e^{\sigma/q_i^*} + e^{-\sigma/q_i^*})^2} \leq 1. \tag{18}$$

Furthermore, if $\sigma_0 = 0$ then (17) can be expressed as

$$h_i(\sigma) = \frac{\partial h_i(\bar{\sigma})}{\partial \sigma}\sigma \tag{19}$$

Let there exists a positive constant h^* satisfies $0 < h_i^* \leq 1$.

Now the transformed dynamics can be written as:

$$\begin{aligned} \dot{x}_1 &= x_2 + h_1(\sigma_1) \\ \dot{x}_2 &= \Theta_1^\top(z)\Phi_1 + b_p h_p(u_p) - \dot{h}_1(\sigma_1) - \ddot{z}_{d1} \\ \dot{x}_3 &= x_4 + h_2(\sigma_2) \\ \dot{x}_4 &= \Theta_2^\top(z)\Phi_2 + b_y h_y(u_y) - \dot{h}_2(\sigma_2) - \ddot{z}_{d3} \end{aligned} \tag{20}$$

We propose following prescribed-time control law for the system (20)

$$u_p = \hat{\rho}_1 \bar{u}_p \tag{21}$$
$$u_y = \hat{\rho}_2 \bar{u}_y \tag{22}$$

with

$$\bar{u}_p = -a_1^* x_1 - a_1^* \Theta_1^\top(z)\hat{\Phi}_1 - \gamma_2 \varphi(t,T_p) x_2 + \dot{h}_1(\sigma_1) + \ddot{z}_{d1}$$
$$\bar{u}_y = -a_2^* x_3 - a_2^* \Theta_2^\top(z)\hat{\Phi}_2 - \gamma_4 \varphi(t,T_p) x_4 + \dot{h}_2(\sigma_2) + \ddot{z}_{d3}$$

The updating law for the unknown parameters are chosen as

$$\dot{\hat{\Phi}}_1 = \Xi_1 \Theta_1 x_2 \tag{23}$$
$$\dot{\hat{\Phi}}_2 = \Xi_2 \Theta_2 x_4 \tag{24}$$
$$\dot{\hat{\rho}}_1 = -\Upsilon_1 \bar{u}_p x_2 \tag{25}$$
$$\dot{\hat{\rho}}_1 = -\Upsilon_2 \bar{u}_y x_4 \tag{26}$$

where, $a_1^* \geq \frac{1}{h_p^*}$, $a_2^* \geq \frac{1}{h_y^*}$, γ_2, γ_4, Υ_1 and Υ_2 are the positive design constants, Ξ_1 and Ξ_2 are the adaptive gain positive definite matrices, $\hat{\Phi}_1$, $\hat{\Phi}_2$, $\hat{\rho}_1$ and $\hat{\rho}_2$ are the estimates of Φ_1, Φ_2, $\rho_1 = \frac{1}{b_p}$ and $\rho_2 = \frac{1}{b_y}$ respectively. Let $\tilde{\Phi}_1 = \Phi_1 - \hat{\Phi}_1$, $\tilde{\Phi}_2 = \Phi_2 - \hat{\Phi}_2$, $\tilde{\rho}_1 = \rho_1 - \hat{\rho}_1$ and $\tilde{\rho}_2 = \rho_2 - \hat{\rho}_2$ be the parameter estimation error.

Theorem 1. For the nonlinear system (20) along with Assumption 1 and Assumption 2, under the proposed control law (21) and (22) with the parameter updating law (23), (24), (25) and (26), the following conclusions can be drawn:

- All signals of the closed loop system (20) are bounded.
- The prescribed-time tracking of reference signals is achieved, i.e. $\lim_{t\to T_p}\big[z_1 - z_{d1}\big] = \lim_{t\to T_p} x_1 = 0$, $\lim_{t\to T_p}\big[z_3 - z_{d3}\big] = \lim_{t\to T_p} x_3 = 0$.

Proof. Late us choose the candidate Lyapunov function as

$$V = V_1 + V_2 \tag{27}$$

with

$$V_1 = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}x_3^2 + \frac{1}{2}x_4^2$$
$$V_2 = \frac{1}{2}\tilde{\Phi}_1^\top \Xi_1^{-1}\tilde{\Phi}_1 + \frac{1}{2}\tilde{\Phi}_2^\top \Xi_2^{-1}\tilde{\Phi}_2 + \frac{b_p}{2\Upsilon_1}\tilde{\rho}_1^2 + \frac{b_y}{2\Upsilon_2}\tilde{\rho}_2^2$$

Now, taking the time derivative of V along the solution to (20), we get

$$\begin{aligned}
\dot{V} &= x_1\dot{x}_1 + x_2\dot{x}_2 + x_3\dot{x}_3 + x_4\dot{x}_4 \\
&\quad - \tilde{\Phi}_1^\top \Xi_1^{-1}\dot{\hat{\Phi}}_1 - \tilde{\Phi}_2^\top \Xi_2^{-1}\dot{\hat{\Phi}}_2 - \frac{b_p}{\Upsilon_1}\tilde{\rho}_1\dot{\hat{\rho}}_1 - \frac{b_y}{\Upsilon_2}\tilde{\rho}_2\dot{\hat{\rho}}_2 \\
&= x_1(x_2 + h_1(\sigma_1)) \\
&\quad + x_2\Big(\Theta_1^\top(z)\Phi_1 + b_p h_p(u_p) - \dot{h}_1(\sigma_1) - \ddot{z}_{d1}\Big) \\
&\quad + x_3(x_4 + h_2(\sigma_2)) \\
&\quad + x_4\Big(\Theta_2^\top(z)\Phi_2 + b_y h_y(u_y) - \dot{h}_2(\sigma_2) - \ddot{z}_{d2}\Big) \\
&\quad - \tilde{\Phi}_1^\top \Xi_1^{-1}\dot{\hat{\Phi}}_1 - \tilde{\Phi}_2^\top \Xi_2^{-1}\dot{\hat{\Phi}}_2 - \frac{b_p}{\Upsilon_1}\tilde{\rho}_1\dot{\hat{\rho}}_1 - \frac{b_y}{\Upsilon_2}\tilde{\rho}_2\dot{\hat{\rho}}_2
\end{aligned} \tag{28}$$

Moreover, by using (19), we have

$$x_1 h_1(\sigma_1) = x_1 \frac{\partial h_1(\bar{\sigma}_1)}{\partial \sigma_1}\sigma_1 \leq -h_1^* \gamma_1 \varphi(t,t_p) x_1^2 \tag{29}$$

$$\begin{aligned}
x_2 b_p h_p(u_p) &= x_2 \frac{\partial h_p(\bar{u}_p)}{\partial u_p} u_p \leq h_p^* b_p \hat{\rho}_1 x_2 \Big(-a_1^* x_1 \\
&\quad - a_1^*\Theta_1^\top(z)\hat{\Phi}_1 - \gamma_2\varphi(t,T_p)x_2 + \dot{h}_1(\sigma_1) + \ddot{z}_{d1}\Big)
\end{aligned} \tag{30}$$

$$x_3 h_2(\sigma_2) = x_3 \frac{\partial h_2(\bar{\sigma}_2)}{\partial \sigma_2}\sigma_2 \leq -h_2^* \gamma_3 \varphi(t,t_p) x_3^2 \tag{31}$$

$$\begin{aligned}
x_4 b_y h_y(u_y) &= x_4 \frac{\partial h_y(\bar{u}_y)}{\partial u_y} u_y \leq h_y^* b_y \hat{\rho}_2 x_4 \Big(-a_2^* x_3 \\
&\quad - a_2^*\Theta_2^\top(z)\hat{\Phi}_2 - \gamma_4\varphi(t,T_p)x_4 + \dot{h}_2(\sigma_2) + \ddot{z}_{d3}\Big)
\end{aligned} \tag{32}$$

Substituting (29)-(32) into (28), we obtain

$$\begin{aligned}
\dot{V} \leq &- h_1^*\gamma_1\varphi(t,t_p)x_1^2 - h_p^*\gamma_2\varphi(t,t_p)x_2^2 - h_2^*\gamma_3\varphi(t,t_p)x_3^2 \\
&- h_y^*\gamma_4\varphi(t,t_p)x_4^2 + \Theta_1^\top\tilde{\Phi}_1 x_2 + \Theta_2^\top\tilde{\Phi}_2 x_4 - b_p\tilde{\rho}_1\bar{u}_p x_2 \\
&- b_y\tilde{\rho}_2\bar{u}_y x_4 - \tilde{\Phi}_1^\top\Xi_1^{-1}\dot{\hat{\Phi}}_1 - \tilde{\Phi}_2^\top\Xi_2^{-1}\dot{\hat{\Phi}}_2 \\
&- \frac{b_p}{\Upsilon_1}\tilde{\rho}_1\dot{\hat{\rho}}_1 - \frac{b_y}{\Upsilon_2}\tilde{\rho}_2\dot{\hat{\rho}}_2
\end{aligned} \tag{33}$$

Further simplifying (33), one can get

$$\begin{aligned}
\dot{V} \leq &- h_1^*\gamma_1\varphi(t,t_p)x_1^2 - h_p^*\gamma_2\varphi(t,t_p)x_2^2 - h_2^*\gamma_3\varphi(t,t_p)x_3^2 \\
&- h_y^*\gamma_4\varphi(t,t_p)x_4^2 - \tilde{\Phi}_1^\top\Xi_1^{-1}\Big(\dot{\hat{\Phi}}_1 - \Xi_1\Theta_1 x_2\Big) \\
&- \tilde{\Phi}_2^\top\Xi_2^{-1}\Big(\dot{\hat{\Phi}}_2 - \Xi_2\Theta_2 x_4\Big) - \tilde{\rho}_1\frac{b_p}{\Upsilon_1}\Big(\dot{\hat{\rho}}_1 + \Upsilon_1\bar{u}_p x_2\Big) \\
&- \tilde{\rho}_2\frac{b_y}{\Upsilon_2}\Big(\dot{\hat{\rho}}_2 + \Upsilon_2\bar{u}_y x_4\Big)
\end{aligned} \tag{34}$$

Since, the parameters update law (23), (24), (25) and (26), cancel the last four terms of the equation (34). Thus, we have

$$\begin{aligned}
\dot{V} \leq &- h_1^*\gamma_1\varphi(t,t_p)x_1^2 - h_p^*\gamma_2\varphi(t,t_p)x_2^2 - h_2^*\gamma_3\varphi(t,t_p)x_3^2 \\
&- h_y^*\gamma_4\varphi(t,t_p)x_4^2 \\
\leq &- \gamma\varphi(t,t_p)V_1
\end{aligned} \tag{35}$$

where $\gamma = \min\{2h_1^*\gamma_1, 2h_p^*\gamma_2, h_2^*\gamma_3, h_y^*\gamma_4\}$.

The prescribed-time stabilization of 2-DOF helicopter follows from Lemma 1. Utilizing the LaSalle-Yoshizawa theorem (Khalil (2002)), the boundedness of V can be ensured. This implies that x_1, x_2, x_3 and x_4 are bounded and prescribed-time stable and x_1, x_2, x_3, $x_4 \to 0$ as $t \to T_p$, also $\hat{\Phi}_1$ and $\hat{\Phi}_2$ are bounded. Since $x_1 = z_1 - z_{d1}$ and $x_3 = z_3 - z_{d3}$, hence the desired reference signals tracking is also obtained in prescribed-time T_p, hence, the upper bound of the settling-time can be prescribed. Moreover, z_1 and z_3 are also bounded since x_1 and x_3 are bounded and since z_{d1} and z_{d3} are bounded by Assumption 2. From (14) and (15) it follows that the virtual controls σ_1 and σ_2 are also bounded. Similarly, the boundedness of the control inputs can be guaranteed from (21) and (22).

4. RESULT DISCUSSION

Using MATLAB/Simulink and a Quanser 2-DOF helicopter, the performance, and efficacy of the proposed prescribed-time adaptive backstepping control law are demonstrated in this section. Angles and angular velocities initial conditions are both set to zero, while a fixed sample

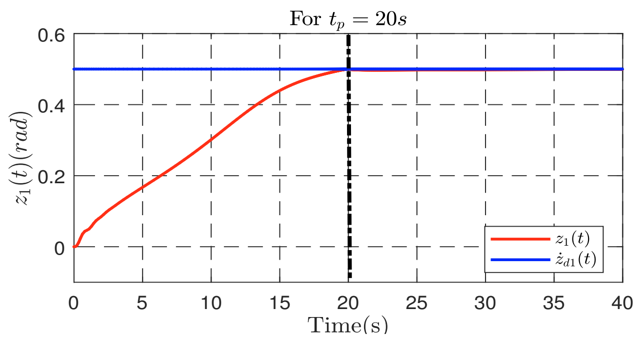

Fig. 2. Pitch angle (z_1) reference tracking for $T_p = 20s$

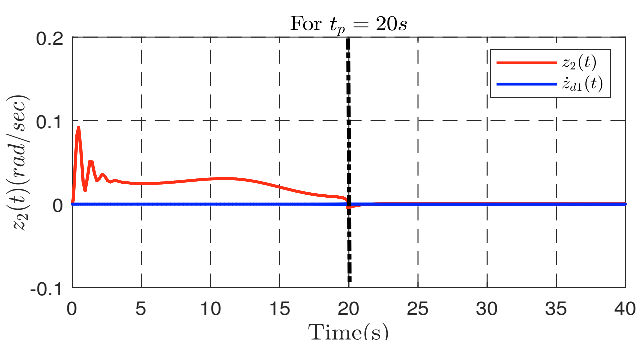

Fig. 3. Pitch velocity (z_2) for $t_p = 20s$

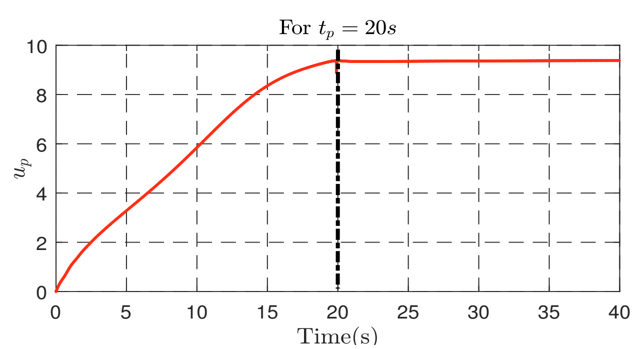

Fig. 4. Control input u_p for $t_p = 20s$

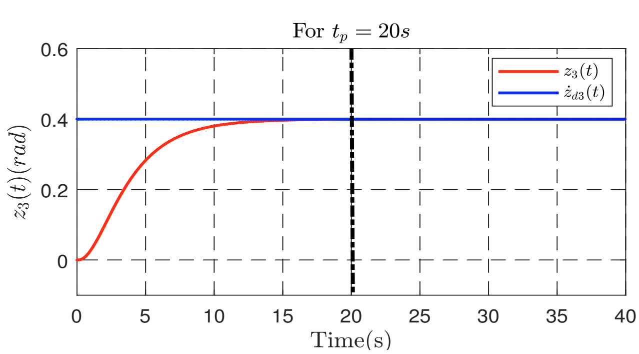

Fig. 5. Yaw angle (z_3) reference tracking for $t_p = 20s$

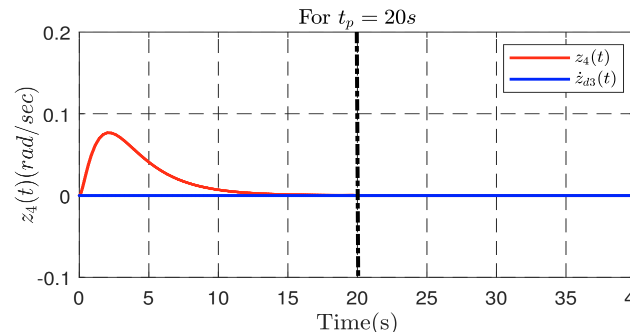

Fig. 6. Yaw velocity (z_4) for $t_p = 20s$

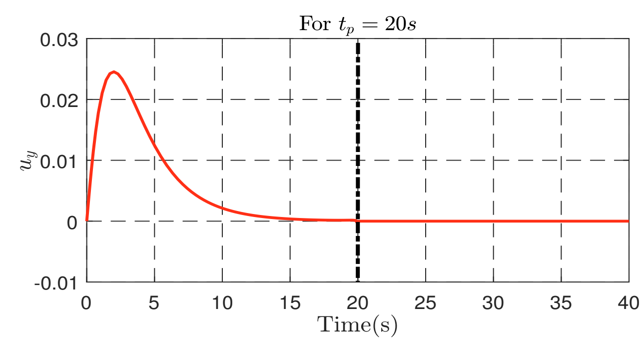

Fig. 7. Control input u_y for $t_p = 20s$

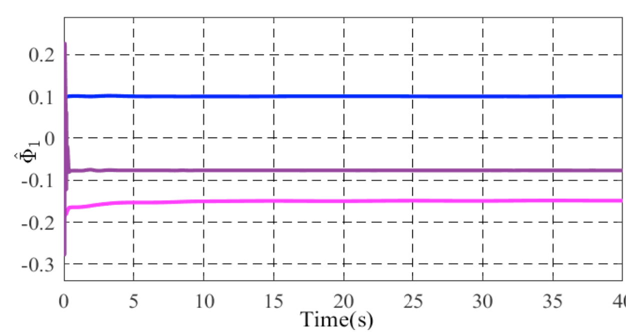

Fig. 8. Estimate of the uncertain parameter vector $\hat{\Phi}_1$

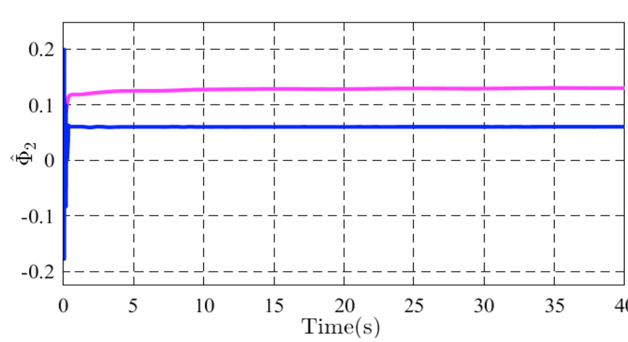

Fig. 9. Estimate of the uncertain parameter vector $\hat{\Phi}_2$

duration of 0.001s is employed. The values of the design constants $\gamma_1, \gamma_2, \gamma_3$, and γ_4 and the aptative law gains Ξ_i and Υ_i is found by trial and error method. The simulations are performed considering the desired time of convergence as $T_p = 20s$ and initial time $t_0 = 0s$. A constant reference with amplitude of 0.5 radian and 0.4 radian is applied to the pitch angle and yaw angle respectively.

The results from simulation on the Quanser 2-DOF with the proposed prescribed-time adaptive controller is shown in Fig. (2)-(7). Fig (2) shows that the desired pitch angle tracking is achieved in prescribed-time $T_p = 20s$. The corresponding pitch angular velocity is shown in Fig. (3). Similarly, for prescribed-time $T_p = 20s$, the required input voltage plot is shown in Fig. (4). The yaw angle tracking is achieved in Fig. (5). In Fig. (6) and Fig. (7) the angular velocity of yaw angle and required input voltage are shown respectively. Fig. (8) and Fig. (9) shows the estimates of $\hat{\Phi}_1$ and $\hat{\Phi}_2$. From the obtained results, one can conclude that the convergence is achieved within a priory chosen time. Moreover, all the closed-loop signals of the system are bounded.

Remark 3. Since the obtained control law is applicable only till the prescribed time T_p, Ensuring the tracking after T_p, it is necessary to have control input throughout the operation of the considered system. Here, in this case, for all $t > T_p$, we applied the control law as given in (Ahn et al. (2013)).

5. CONCLUSION

This paper proposes a constraint prescribed-time adaptive backstepping controller for an uncertain 2-DOF helicopter that tracks pitch and yaw angle reference trajectory in the prescribed time. At first, the Euler-Lagrange equations are utilized to obtain the mathematical model of the 2-DOF helicopter. A constraint function is also employed to limit the virtual control as the time approaches the terminal time. Using candidate Lyapunov functions, a theoretical demonstration of prescribed-time stabilization with adaptive backstepping is provided, guaranteeing the boundedness of all signals in the closed loop system and attitude tracking in the prescribed time. Finally, simulations are carried out to show the effectiveness and control potential of the designed control scheme. As a future scope of this work, the proposed method can be extended for the other benchmark setups and control with state and input quantization.

REFERENCES

Adetola, V. and Guay, M., Finite-time parameter estimation in adaptive control of nonlinear systems. *IEEE Transactions on Automatic Control*, 53(3), pp.807-811, 2008.

Ahn, K.K., Nam, D.N.C. and Jin, M., Adaptive backstepping control of an electrohydraulic actuator. *IEEE/ASME transactions on mechatronics*, 19(3), pp.987-995, 2013.

Al Issa, S. and Kar, I., Event-triggered Adaptive Backstepping Control of Nonlinear Uncertain Systems with Input Delay. *IFAC-PapersOnLine*, 55(1), pp.667-672, 2022.

Åström, K.J., Theory and applications of adaptive control—a survey. automatica, 19(5), pp.471-486, 1983.

Bhat, S.P. and Bernstein, D.S., Finite-time stability of continuous autonomous systems. *SIAM Journal on Control and optimization*, 38(3), pp.751-766, 2000.

Boubakir, A., Labiod, S., Boudjema, F. and Plestan, F., Design and experimentation of a self-tuning pid control applied to the 3dof helicopter. *Archives of Control Sciences*, (3), 2013.

Chatterjee, K., Singh, V.K., Prasun, P., Kamal, S., Ghosh, S. and Dinh, T.N., Fixed-time event-triggered control under denial-of-service attacks. *European Journal of Control*, 69, p.100765, 2023.

Chen, F., Wu, Q., Jiang, B. and Tao, G., A reconfiguration scheme for quadrotor helicopter via simple adaptive control and quantum logic. *IEEE Transactions on Industrial Electronics*, 62(7), pp.4328-4335, 2015.

Ding, J. and Zhang, W., Finite-time adaptive control for nonlinear systems with uncertain parameters based on the command filters. *International Journal of Adaptive Control and Signal Processing*, 35(9), pp.1754-1767, 2021.

Hua C., Ning P., and Li K., Adaptive Prescribed-Time Control for a Class of Uncertain Nonlinear Systems. *IEEE Transactions on Automatic Control.*, vol. 67, no. 11, pp. 6159-6166, 2022.

Joelianto, E., Sumarjono, E.M., Budiyono, A. and Penggalih, D.R., Model predictive control for autonomous unmanned helicopters. *Aircraft Engineering and Aerospace Technology.*, 2011.

Khalil, H.K., Nonlinear systems third edition. *Patience Hall*, 115, 2002.

Krstic, M., Kokotovic, P.V. and Kanellakopoulos, I., Nonlinear and adaptive control design. *John Wiley and Sons*, Inc., 1995.

Marconi, L. and Naldi, R., Aggressive control of helicopters in presence of parametric and dynamical uncertainties. *Mechatronics*, 18(7), pp.381-389, 2008.

Pal, A.K., Kamal, S., Nagar, S.K., Bandyopadhyay, B. and Fridman, L., Design of controllers with arbitrary convergence time. *Automatica*, 112, p.108710, 2020.

Pandey, S.K. and Laxmi, V., Optimal control of twin rotor MIMO system using LQR technique. *Computational Intelligence in Data Mining*, Volume 1 (pp. 11-21), Springer, New Delhi, 2015.

Polyakov, A., Nonlinear feedback design for fixed-time stabilization of linear control systems. *IEEE Transactions on Automatic Control*, 57(8), pp.2106-2110, 2011.

Singh, V.K., Prasun, P., Singh, B., Kamal, S. and Ghosh, S. October. Neural Network Control based Stabilization of Nonlinear Systems in Arbitrary Time. *47th Annual Conference of the IEEE Industrial Electronics Society*, pp. 1-6, 2021.

Singh, V. K., Prasun, P., Chatterjee, K., Kamal, S., Ghosh, S. and Bandyopadhyay B. Optimal Sliding Mode Control with Predefined Upper Bound of Settling Time. *16th International Workshop on Variable Structure Systems (VSS)*, pp. 260-265, 2022.

Song, Y., Wang, Y., Holloway, J. and Krstic, M., Time-varying feedback for regulation of normal-form nonlinear systems in prescribed finite time. *Automatica*, 83, pp.243-251, 2017.

Wang, W., Zhou, J., Wen, C. and Long, J., Adaptive backstepping control of uncertain nonlinear systems with input and state quantization. *IEEE Transactions on Automatic Control*, 2021.

Wen, C., Zhou, J., Liu, Z. and Su, H., Robust adaptive control of uncertain nonlinear systems in the presence of input saturation and external disturbance. emphIEEE Transactions on Automatic Control, 56(7), pp.1672-1678, 2011.

Zhang, Z., Wen, C., Xing, L. and Song, Y., Adaptive Output Feedback Control of Nonlinear Systems with Mismatched Uncertainties under Input/Output Quantization. *IEEE Transactions on Automatic Control*, 2022.

Zhou, J., Wen, C. and Zhang, Y., Adaptive backstepping control of a class of uncertain nonlinear systems with unknown backlash-like hysteresis. *IEEE transactions on Automatic Control*, 49(10), pp.1751-1759, 2004.

Zou, Y. and Huo, W., Adaptive tracking control for a model helicopter with disturbances. *In 2015 American Control Conference (ACC)* , (pp. 3824-3829), July. 2015.